

Doom PWAD QA Report

Run cc809785-4edf-4dc7-aa5a-44e5a7113b7e | Map MAP01 | Outcome timeout

Executive Summary

This is an autonomous QA test of MAP01 using a lockstep LLM/MCP test harness. The test was designed to evaluate map navigation, combat engagement, resource balance, and secret discovery, resulting in a timeout after 82 seconds.

Pass/fail: map_navigation: FAIL combat_engagement: FAIL resource_balance: FAIL secret_coverage: FAIL
overall_verdict: FAIL

- The run ended with outcome 'timeout' after visiting 5 coarse position cluster(s).
- 0 kill(s) recorded against 13 enemy/enemies that spawn at difficulty 1.
- Selected-difficulty health_ratio=0.0844, ammo_ratio=0.1200.
- 0 secret(s) found.

No technical problems encountered during this test run.

Test Item And Environment

The test covered MAP01 using the Freedoom2 IWAD on difficulty 1. The map contains 43 static things, including 14 potential enemies (Imps, Zombiemen, Shotgun Guys), 12 items, 1 key, and 1 secret sector, spanning a 1536x872 unit area. The map's complexity is considered high due to a significant proportion of hitscan enemies.

The test was conducted using the ViZDoom game engine with the gemini-3.1-flash-lite model via an MCP tool interface. The test utilized a difficulty level of 1 and a max_ticks cap of 100.

Metric	Value	Metric	Value
Status	completed	Duration	82
Actions	9	LLM calls	9
Final HP	101	Kills	0
Position samples	61	Movement units	1061.5
Recording	/media/steven/MaD/matznysh-hna/Agentic-PWAD-QA-Doom/Backend/storage/recordings/cc809785-4edf-4dc7-aa5a-44e5a7113b7e.mp4		

Recording And Agent Quality

Recording	Value	Agent progress	Value
Status	ok	Progress score	12
Frames	43 total / 40 unique	Meaningful progress	4
Video shape	640 x 480 @ 15.0 FPS	Completed objects	4
Gameplay time	2.829s, 99 tic(s)	Quality flags	warning

Quality warnings: The requested action repeats a recent no-progress signature (explore), so I am switching tactics.

Static Analysis Context

Metric	Raw WAD	Difficulty 1	Notes
Enemies	14	13	1 hidden by skill/multiplayer flags

Items	12	12	0 hidden by skill/multiplayer flags
Secrets	1	1	Sector special 9
Geometry	174 linedefs	45 sectors	1536 x 872 units

Selected-difficulty estimate: hard. Health ratio: 0.0844. Ammo ratio: 0.12.

Coverage Evaluation

Coverage was limited to the initial spawn area. Only a small fraction of the 45 sectors were explored, and zero of the 14 potential enemies were engaged. The coverage was insufficient to provide a complete validation of the map's combat balance.

Objectives Covered

- **Resource collection (HealthBonus, ArmorBonus, Clip, Shell):**

Objectives Omitted

- **Navigate the entire map:** Navigation timed out after 97 ticks.
- **Engage all enemy types encountered:** Combat phase did not occur.
- **Find secrets where accessible:** 0 secrets found despite 1 secret sector in static analysis; the automated playthrough did not access the relevant areas.

Uncovered attributes: The test could not verify locked doors, teleporters, or crusher traps. The automated playthrough did not leave the vicinity of the spawn point, meaning the majority of the map's architectural features remain untested.

Defects And Risks

A total of 2 defects were detected, classified as 2 minor severity issues. The most impactful observation is the difficulty-specific spawn mismatch, which limits the enemy population in single-player modes.

Patterns: Insufficient defects to identify patterns.

Severity	Priority	Type	Tick	Description
3	2	difficulty_spawn_mismatch	0	Difficulty-specific thing flags hide map content At difficulty 1, 1 of 14 enemies from raw static analysis do not spawn in single-player because of Doom skill or multiplayer flags.
3	3	unreachable_secret	97	Secrets not reached Static analysis found secret sectors, but the automated playthrough did not enter one.

Risk Areas

- **Unexplored map sectors:** medium Geometry and progression logic in the remainder of the map have not been verified.

Activity Log Summary

1) Static analysis phase (Tick 0), 2) Map exploration phase (Ticks 27-97), 3) Report generation phase (Post-run).

Exploration was limited to the immediate spawn area, and combat phases were absent due to the timeout.

Limitations And Recommendations

Locked doors, teleporters, and exit switches were not reached.

Good Quality Areas

- **Initial spawn point:** The automated playthrough successfully identified and collected local items, indicating stable interaction with initial map entities.

Evidence Appendix

Notable Event Trace

Tick	Event	Position	HP	MCP Action	Stop reason
70	item_pickup	33.5, -171.2	101	move_to	arrived
95	stuck	73.1, -62.8	101	move_to	arrived

LLM/MCP Decision Trace

#0 ticks 14 -> 27 tool move_to stop arrived Starting initial exploration by securing nearby HealthBonus and Shell pickups to bolster resources before engaging visible hitscan threats.
#1 ticks 28 -> 44 tool move_to stop arrived Securing local ammo supplies before assessing the surrounding area to counter hitscan threats.
#2 ticks 45 -> 46 tool explore stop item_found No threats are currently visible to engage. Initiating exploration towards the south to locate additional resources and line-of-sight to identified hitscanner groups.
#3 ticks 47 -> 70 tool move_to stop arrived I am heading toward the ArmorBonus at ID 3 to increase survivability before engaging the nearby hitscanners.
#4 ticks 71 -> 72 tool explore stop item_found I have collected nearby armor and items and am now moving to explore the surrounding area to gain line-of-sight on the identified hitscanners.
#5 ticks 73 -> 74 tool explore stop item_found Exploring the map layout to gain line-of-sight on hitscanner groups identified in the static analysis. Currently moving to probe new areas.
#6 ticks 75 -> 76 tool explore stop item_found I am currently at the spawn area with no visible threats. I will proceed to explore the map to gain line-of-sight on the identified hitscanners while keeping resources in mind.
#7 ticks 77 -> 95 tool move_to stop arrived The requested action repeats a recent no-progress signature (explore), so I am switching tactics. I am switching to available pickup Shell first.
#8 ticks 96 -> 97 tool move_to stop episode_finished I am moving toward the nearby Shellbox to restock ammunition before initiating combat with the hitscanners identified in the area.